

Reconstructing projective frames from their continuation graphs

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Abstract

Four points with no three collinear in the projective plane $\text{PG}(2, q)$ determine a graph on the points that can be adjoined without creating three collinear points; two vertices are adjacent when they cannot be adjoined together. For $q \geq 9$, we prove that every isomorphism between these uncoloured graphs extends uniquely to a semilinear equivalence of the frames. The graph also determines q . The proof recovers the legal points on each tangent line, groups these lines by their selected point, then uses compatibility between multiplication and translation to recover the field action. Completing a punctured division table makes the reconstruction algorithmic and gives polynomial-time recognition over a supplied finite field, with an independently checkable coordinate certificate. An exhaustive finite computation closes the range $5 \leq q < 9$: semilinear rigidity holds at $q = 7$, whereas $q = 5, 8$ each have exactly two resolutions into four parallel classes of cliques, and the semilinear subgroup has index two in the full automorphism group. The stable-range proof is independent of this computation.

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1 Introduction

Choose four points in a finite projective plane, with no three on a line. Keep the points that can be added without creating a collinear triple, and join two of them when they cannot both be added. Now erase the plane, the chosen points, and all labels. Must every symmetry of the remaining graph extend to a symmetry of the original plane?

More generally, an *arc* K is a set with no three collinear points. A point outside K is a *legal continuation* if adjoining it preserves this property. The continuation graph G_K joins incompatible pairs of legal continuations. Such a pair lies on a *tangent*, a line meeting K in exactly one point. The graph records the conflict but forgets both the tangent and its selected point. Recovering this lost organization is the first obstacle to extending a graph symmetry to the plane.

The selected point on a tangent is its *centre*; a secant meets K in two points. A tangent trace consists of the legal points on that line. The pencil with centre t consists of all lines through t ; its nonempty tangent traces partition the continuation points. We call this partition the class with centre t . We write $\text{PG}(2, q)$ for the plane over $\mathbb{F}_q$. A projectivity is induced by an invertible linear map; allowing a field automorphism as well gives a semilinear collineation. These form $\text{PTL}(3, q)$. Stabilizers of point sets are taken setwise.

A graph symmetry initially preserves only incompatible pairs. To extend it, we recover the tangent traces, group them by centre, and then constrain the field action. The fundamental

theorem of projective geometry applies after incidence is known; it does not recover that incidence from a graph.

Our main result settles the extension question at the smallest arc size for which semilinear rigidity holds uniformly in the field order. Every four-arc in $\text{PG}(2, q)$ is a projective frame.

Theorem 1.1 (Semilinear rigidity). *Let K be a projective frame in $\text{PG}(2, q)$, where $q = p^e \geq 9$. Then, as permutation groups on the legal continuation set,*

$$\text{Aut}(G_K) = \text{Stab}_{\text{PGL}(3, q)}(K), \quad \text{Aut}(G_K) \cong S_4 \times C_e, \quad |\text{Aut}(G_K)| = 24e.$$

Corollary 1.2 (Reconstruction and extension). *Let $K \subset \text{PG}(2, q)$ and $J \subset \text{PG}(2, q')$ be projective frames, where $q, q' \geq 9$. Every graph isomorphism $f : G_K \rightarrow G_J$ forces $q = q'$ and extends uniquely to a semilinear collineation carrying K to J . Thus the abstract graph determines the plane and its frame up to semilinear equivalence within this family.*

The corollary controls individual maps: every relabelling preserving incompatibility has exactly one semilinear extension. Section 7 makes the reconstruction effective: given an arbitrary graph and a finite field of order at least 9, it either rejects or supplies field coordinates whose pairwise comparisons verify the representation. The input is only one uncoloured binary relation, without tangent colours or ambient coordinates. The proof has two stages. First, a geometric bound of five for cliques spread across tangents recovers the tangent traces and a second large-clique argument recovers their four centre classes. Second, after normalizing the frame, the graph becomes the agreement graph of the four maps

$$(x, y) \mapsto x, \quad y, \quad x/y, \quad (x-1)/(y-1).$$

A graph automorphism preserving the recovered classes induces permutations of the four alphabets. Compatibility with both quotient maps forces those permutations to be one field automorphism.

For a concrete model, work over $\mathbb{F}_{13}$. The legal points $(2, 3)$ and $(2, 4)$ have four-coordinate words $(2, 3, 5, 7)$ and $(2, 4, 7, 9)$. They are adjacent because their first coordinates agree: they lie on the same vertical tangent. Agreement means equality in the same position, not merely a shared field value. The task is to recover these coordinate positions when only adjacency remains.

At this order each tangent trace is a ten-vertex clique, whereas a clique spread across several tangents has at most five vertices. Once the traces are recovered, the eleven traces through one centre are pairwise disjoint; a disjointness clique using several centres has at most four members. These two size gaps give the intrinsic reconstruction

$$\text{graph} \longrightarrow \text{tangent traces} \longrightarrow \text{centre classes} \longrightarrow \text{field coordinates}.$$

The last arrow uses both quotient maps: multiplication alone leaves extra symmetries, and the translated quotient removes them.

Related work. Recovering coordinate partitions from agreement graphs has a close Latin-square precedent. Bailey, Cameron, Praeger and Schneider [1, Propositions 2.4 and 2.6] recover the row, column and symbol partitions of a Latin square of order greater than four from large cliques and disjointness, and identify its graph automorphisms with autoparatopisms (permutations of the three positions and their symbols preserving the table). Our triangle graph is the Latin-square graph of a division table. For a frame the table is punctured and a fourth, coupled quotient is present. We recover those four partitions and prove that their compatibility forces a field automorphism, thereby extending each individual graph isomorphism.

One can also represent a vertex by its four tangent lines and regard the graph as the intersection graph of these four-element sets. The standard multipartite argument in the proof of Francetić, Herke, McKay and Wanless [2, Lemma 2.2] bounds non-star cliques by nine; Lemma 3.3 records its general form. Here a star means a family sharing one tangent symbol. Bhattacharya, Godinho, Majumder and Singhi [3, Lemmas 2.3–2.4] give a general bounded-pair-degree reconstruction argument, whose corresponding four-element threshold is fourteen. The frame geometry improves the non-star bound to five (Lemma 3.1), so traces of size six already suffice. Recovering the centre classes and enforcing the two quotient constraints then completes the extension proof.

The four-coordinate model also gives an isometry-extension question for a nonlinear code. Hamming distance counts the positions in which two words differ. Here distinct words have distance three or four, and an edge records distance three, so graph automorphisms are exactly the code’s distance-preserving permutations. In the algebraic setting, Dyshko studies extension of additive isometries of additive maximum distance separable (MDS) codes [6, Sections 1–2]. Here the code has $Q(Q-1)$ words over $Q = q-2$ symbols and is not MDS, whose Singleton bound here is Q^2 . Corollary 6.3 states the ambient Hamming-space extension explicitly; coordinate recovery is part of its proof.

A related finite-geometric question asks when a graph built from field invariants admits symmetries beyond its prescribed geometric action. Gardiner, Praeger and Zhou determine automorphism groups and isomorphisms for cross-ratio graphs on ordered distinct projective-line pairs [7, Theorems 7.1–7.2]. Those graphs prescribe cross-ratio orbits as adjacency. Here adjacency merges four fibre relations, and the proof recovers their separate roles before identifying the field action. The configuration-space interpretation of these fibres is optional background in Section 9.

Boundary and reading guide. The lower boundary is genuine. One-point continuation graphs are disjoint unions of cliques, every two-point continuation graph is a rook graph independent of the plane, and triangles generally admit automorphisms from the multiplicative group of the field that are not semilinear. At $q = 5$ and $q = 8$ the ambient group has index two in the graph automorphism group. The finite classification in Section 8 proves rigidity at $q = 7$. Thus $q \geq 9$ is the optimal uniform cutoff, with a computation-free proof throughout that range. The finite boundary locates exactly where the encoding loses uniqueness: at $q = 5, 8$ the graph admits two resolutions into four parallel classes, exchanged by its extra symmetries. This distinguishes a failure of reconstruction from a limitation of the large-clique method.

For the main theorem, retain the recovery statements in Sections 3–4, then follow the four-coordinate model and the two algebraic lemmas in Section 6. The general k -dependent estimates can be read after the frame case; Section 5 explains why three points do not suffice uniformly. Sections 7 and 8 give, respectively, the recognition algorithm and the finite computational boundary.

2 Continuation graphs and their intersection representation

Let $\Pi = (P, \mathcal{L})$ be a finite projective plane of order q , and let $K \subset P$ be a k -arc. Define

$$V_K = \{x \in P \setminus K : K \cup \{x\} \text{ is an arc}\}.$$

All graphs are simple; adjacency formulas concern distinct vertices. We write $N_G(v)$ for the open neighbourhood of v in G , omitting the subscript when the graph is clear. The

continuation graph G_K has vertex set V_K , with

$$x \sim_K y \iff \text{some line contains } x, y, \text{ and one point of } K. \quad (1)$$

Because x and y are individually legal, a line witnessing (1) contains exactly one point of K . Thus every edge has a unique *selected centre*.

For $t \in K$, let Σ_t be the $q + 2 - k$ tangent lines through t . A hypergraph is a family of subsets, called hyperedges. It is k -uniform if each hyperedge has k elements, and linear if distinct hyperedges meet in at most one element. Its line graph joins intersecting hyperedges. For $x \in V_K$, put

$$S_x = \{xt : t \in K\} \subseteq \prod_{t \in K} \Sigma_t.$$

Proposition 2.1 (Intersection representation). *Suppose $k \geq 2$. The sets S_x form a k -uniform linear hypergraph $\mathcal{H}_K$, and for distinct $x, y \in V_K$,*

$$|S_x \cap S_y| = \begin{cases} 1, & x \sim_K y, \\ 0, & x \not\sim_K y. \end{cases}$$

Consequently G_K is the line graph of $\mathcal{H}_K$.

Proof. Since x is legal, xt is tangent to K for every $t \in K$, so $|S_x| = k$. Two such sets share a tangent symbol exactly when the corresponding points lie on that tangent and hence are adjacent. Distinct points cannot share two projective lines, proving linearity. Moreover, for two distinct centres t, u , the lines xt and xu are distinct because x is legal, and their intersection is x . Hence S_x determines x , so distinct points give distinct hyperedges. $\square$

For $k \geq 2$, after labelling the tangents in each Σ_t , the map

$$x \mapsto (xt)_{t \in K} \in \prod_{t \in K} \Sigma_t$$

is an injective nonlinear code of length k . Distinct words agree in at most one coordinate. Thus their Hamming distance is $k - 1$ for an adjacent pair and k otherwise. The graph therefore retains the entire distance function on this code, even though it forgets the names of the coordinates and their symbols.

Normalize the frame as

$$K = \{(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1), (1 : 1 : 1)\}.$$

Proposition 2.2 (Four-coordinate normal form). *The legal continuation points are*

$$\Omega = \{(x, y) : x, y \notin \{0, 1\}, x \neq y\}.$$

The map

$$(x, y) \mapsto \left(x, y, \frac{x}{y}, \frac{x-1}{y-1}\right) \quad (2)$$

is injective, and two vertices are adjacent exactly when their images agree in one coordinate.

Proof. The six frame secants have equations $X = 0$, $Y = 0$, $Z = 0$, $X = Y$, $X = Z$, and $Y = Z$. After normalizing $Z = 1$, avoiding them gives the displayed set Ω . The parameters x , y , x/y , and $(x-1)/(y-1)$ label lines through $(0 : 1 : 0)$, $(1 : 0 : 0)$, $(0 : 0 : 1)$, and $(1 : 1 : 1)$, respectively. These are the four pencil parameters in (2). $\square$

3 Intrinsic recovery of tangent traces

For a tangent line ℓ , write $B_\ell = \ell \cap V_K$. This is a clique of G_K , but a clique need not a priori lie on one tangent: different edges may have different selected centres.

Lemma 3.1 (Five-clique bound for a frame). *Let K be a projective frame in $\text{PG}(2, q)$. Every clique not contained in a tangent trace has at most five vertices. This bound is attained at $q = 7$.*

Proof. Let C be such a clique. Each tangent contains at most three points of C : from a point outside it, those points must be reached through distinct members of the other three frame centres. If $|C| \geq 6$, the four centred lines through any vertex distribute at least five other vertices, so some tangent ℓ contains three points of C . At least three lie outside it.

Send ℓ to infinity, write its centre as t , and affinely normalize the other centres to $u = (0, 0)$, $v = (1, 0)$, $w = (0, 1)$. Since K is a frame, t is not the direction of a triangle side. Fix an outside point $P = (a, b)$ and put $c = 1 - a - b$; legality gives $abc \neq 0$. The three ideal clique points are exactly the directions from P to u, v, w . Every other outside point assigns these same directions to the three centres by a permutation. Each assignment has at most one point, as two of its distinct directions already specify an intersection. The identity gives P . Two points can share a line through an affine centre only if their assignments agree there; if they agree nowhere, adjacency requires their joining direction to be t .

We show that no two nonidentity assignments can coexist with P . A transposition of centres r, s has candidate $r + s - P$, by intersecting the two exchanged parallel lines. Candidates for two distinct transpositions differ by a triangle-side vector. Their assignments agree nowhere, so they cannot be adjacent.

The two three-cycle candidates are $R = (c, a)$ and $S = (b, c)$, each requiring $ab + bc + ca = 0$. For example, writing $R = (h, j)$, one cycle gives

$$bh = (a - 1)j, \quad (b - 1)(h - 1) = aj, \quad bh = a(j - 1).$$

These equations give $j = a$, $h = c$ and the stated condition; reversing the cycle interchanges the roles. Both cycles disagree with P at every centre. If both candidates were adjacent to P , then PR and PS would have direction t . This contradicts

$$\det(R - P, S - P) = a^2 + b^2 + c^2 - ab - bc - ca = (a + b + c)^2 - 3(ab + bc + ca) = 1.$$

This identity also holds in characteristics two and three.

Finally, for a transposition and a cycle, relabel the triangle so the transposition swaps v, w . Its candidate is $(1 - a, 1 - b)$; agreement at u forces $a = b$. The cycle candidates are then (c, a) and (a, c) . A distinct such candidate joins $P = (a, a)$ in a triangle-side direction, so it cannot be adjacent to P . Hence at most two clique points lie outside ℓ , a contradiction.

For sharpness, in the coordinates of Proposition 2.2 over $\mathbb{F}_7$, the five points

$$(2, 3), (2, 4), (5, 4), (6, 2), (6, 4)$$

have words $(2, 3, 3, 4)$, $(2, 4, 4, 5)$, $(5, 4, 3, 6)$, $(6, 2, 3, 5)$, $(6, 4, 5, 4)$, respectively. Every pair agrees in one position, and no position is constant on all five. $\square$

Corollary 3.2 (Frame tangent-trace recovery). *For a projective frame in $\text{PG}(2, q)$ with $q \geq 9$, the tangent traces are exactly the maximal cliques with more than five vertices.*

Proof. Every trace has $q - 3 > 5$ vertices by Proposition 2.2. Lemma 3.1 puts every larger clique in a trace. It also makes each full trace maximal, since adjoining an outside vertex would contradict the same bound. $\square$

The following general bound applies to arcs in arbitrary finite projective planes. It is independent of the sharper frame argument.

Lemma 3.3 (Non-tangent clique bound). *Assume $k \geq 2$. If a clique C of G_K is not contained in a tangent trace, then*

$$|C| \leq k(k-2) + 1.$$

Proof. Fix $x \in C$ and partition $C \setminus \{x\}$ into sets C_t , where $y \in C_t$ when the edge xy is centred at $t \in K$. For a nonempty C_t , choose $z \in C$ outside the line xt ; such a z exists because C is not contained in that trace. Let u be the centre of xz .

For $y \in C_t$, let s_y be the centre of zy . The centres s_y are distinct: otherwise two distinct points of xt would also lie on one line through z . Moreover $s_y \notin \{t, u\}$, since either equality forces z or y onto two distinct lines through x . Hence $|C_t| \leq k-2$. There are at most k parts, and therefore $|C| - 1 = \sum_t |C_t| \leq k(k-2)$. $\square$

Deza's weak- Δ -system theorem gives the related generic bound $k^2 - k + 1$ for a non-sunflower family of k -sets with constant pairwise intersection size [5]; in our application that size is one. The multipartite improvement is already implicit in the argument of [2, Lemma 2.2]: a non-star linear intersecting k -partite hypergraph has maximum degree $\Delta \leq k-1$, and counting edges meeting one edge gives $h \leq k(\Delta-1) + 1 \leq (k-1)^2$. These intermediate steps need only the non-star condition, although the stated lemma assumes a stronger covering-number hypothesis. Lemma 3.3 specializes that argument to tangent symbols; it needs no further projective incidence.

Theorem 3.4 (Tangent-trace recovery). *If*

$$q - \binom{k-1}{2} > k(k-2) + 1, \tag{3}$$

then the tangent traces are exactly the maximal cliques of G_K with more than $k(k-2) + 1$ vertices. Hence every abstract graph isomorphism $G_K \rightarrow G_J$, where J is a k -arc whose plane order also satisfies (3), canonically induces an isomorphism of the uncoloured tangent-trace hypergraphs.

Proof. A tangent contains q points besides its centre. At most $\binom{k-1}{2}$ of them lie on secants determined by the other selected points, so every tangent trace has more than the threshold in the statement. Lemma 3.3 says that every clique above that threshold lies in one tangent trace. Such a trace is maximal: a vertex adjacent to its whole trace would create a larger clique and hence would itself lie on the same tangent. Conversely every maximal clique above the threshold is therefore the full trace containing it. $\square$

4 Intrinsic recovery of selected centres

Assume the tangent traces have been recovered, either by Corollary 3.2 or by Theorem 3.4. Form a graph D_K whose vertices are the recovered tangent traces, joining two traces when they are disjoint as subsets of V_K . The selected centres are no longer given. We recover them by the same size-gap principle: traces through one centre form a large disjoint family, whereas a family mixing centres is small.

Theorem 4.1 (Centre-class recovery). *Suppose all tangent traces are nonempty and have been recovered. Put*

$$T = q + 2 - k, \quad c = \binom{k-2}{2}.$$

If $T > kc$, then the k classes of tangent traces having a common selected centre are exactly the maximal cliques of D_K with more than kc members. Thus G_K intrinsically recovers its tangent traces, their partition into selected centres, and every tangent-incidence triple (t, x, y) with $x, y \in V_K$.

Proof. The T traces centred at one $t \in K$ are pairwise disjoint. Fix a trace A centred at t and another centre $u \neq t$. A tangent trace at u can be disjoint from A only when the two ambient tangent lines meet at an illegal point. That point cannot lie on a secant involving t or u ; it must lie on one of the c secants whose endpoints are both in $K \setminus \{t, u\}$. Each such secant determines at most one tangent through u , so at most c traces centred at u are disjoint from A .

Now let a clique of D_K use at least two centre classes, and choose traces A, B from distinct classes. The trace A bounds the contribution from each other class by c , while B gives the same bound for the class of A . The mixed clique therefore has at most kc members. Under $T > kc$, every larger clique lies in one centre class, and maximality recovers the whole class. $\square$

For reference, the resulting sufficient integer thresholds are

k	3	4	5	6	8
q	7	13	23	41	127.

These are the general plane bounds; for Desarguesian frames the entry 13 improves to 9 by Corollary 3.2. The disjointness argument is plane-specific: in higher dimension, tangent lines with different centres may be skew.

5 Why uniform extension needs four points

The number of selected points changes what adjacency can detect. Two centres give only rows and columns. A third adds division, but division alone allows every automorphism of the multiplicative group. The fourth centre will add a translated quotient, linking this multiplicative data to the field's additive structure.

Proposition 5.1 (One point). *For a one-point arc in $\text{PG}(n, q)$, the continuation graph is a disjoint union of $1 + q + \dots + q^{n-1}$ copies of K_q . This structure permits permutations of individual clique fibres; semilinear rigidity does not hold uniformly for one-point arcs.*

Theorem 5.2 (Two points). *If $K = \{a, b\}$ in any projective plane of order q , then*

$$G_K \cong K_q \square K_q.$$

In particular, its isomorphism type is independent of the projective plane.

Proof. The legal points are precisely the q^2 points outside ab . Map such a point x to the ordered pair of tangent lines (ax, bx) . Every choice of one tangent at a and one at b meets in one legal point, and two points are adjacent exactly when they share one of these tangent lines. This is rook adjacency. $\square$

Theorem 5.3 (Three points). *Let K be a triangle in $\text{PG}(2, q)$. With its vertices normalized to the coordinate points, the legal vertices are $(x : y : 1)$ with $x, y \in \mathbb{F}_q^\times$, and*

$$(x, y) \sim (x', y') \iff x = x' \text{ or } y = y' \text{ or } x/y = x'/y'. \quad (4)$$

Every automorphism α of the multiplicative group $\mathbb{F}_q^\times$ therefore induces the graph automorphism $(x, y) \mapsto (\alpha(x), \alpha(y))$. This is not induced by a semilinear collineation unless α is a field automorphism.

Proof. The three equalities in (4) are the three tangent-pencil parameters. A multiplicative-group automorphism preserves equality and quotient. For $q = 2$ the multiplicative group is trivial, so the conclusion holds directly. Assume $q \geq 3$. Each pencil trace then contains at least two legal points and determines its ambient line. If an ambient semilinear collineation induces this map, preservation of the three pencil partitions forces it to fix the triangle vertices. In normalized coordinates it has the form $(x, y) \mapsto (A\sigma(x), B\sigma(y))$ for a field automorphism σ . Since $\alpha(1) = 1$, equality with the displayed graph automorphism forces $A = B = 1$ and $\alpha = \sigma$. $\square$

For example, inversion on $\mathbb{F}_5^\times$ is nonambient. These examples separate intrinsic centre recovery from ambient semilinear extension. The failure is uniform, not absolute: at $q = 3$ the triangle graph is K_4 , and its triangle stabilizer $C_2^2 \rtimes S_3 \cong S_4$ acts faithfully on all four legal points.

6 The four-point frame

The geometric step has recovered four partitions into tangent traces. We now ask which permutations of their labels preserve the graph. Three partitions encode multiplication; the fourth compares multiplication before and after a translation. For a first reading, retain the conclusions of the next two lemmas: the first forces one multiplicative permutation, and the second forces that permutation to preserve the field.

Write $D = \mathbb{F}_q \setminus \{0, 1\}$. For $q \geq 9$, Corollary 3.2 and Theorem 4.1 recover the four coordinate partitions. After composing with the unique projectivity inducing the corresponding permutation of the frame, a graph automorphism preserves all four partitions. Preserving a partition permits relabelling its fibres; write $\alpha, \beta, \gamma, \delta$ for these four permutations of D . The first two send a vertex to $(\alpha(x), \beta(y))$. Its last two labels can also be read from the quotient formulas, so consistency gives

$$\frac{\alpha(x)}{\beta(y)} = \gamma(x/y), \quad (5)$$

$$\frac{\alpha(x) - 1}{\beta(y) - 1} = \delta\left(\frac{x - 1}{y - 1}\right) \quad (6)$$

for all distinct $x, y \in D$. These are initially independent permutations of the row, column and quotient labels. The missing diagonal $x = y$ will force the row and column permutations to coincide.

Lemma 6.1 (Relabelling a punctured division table). *Suppose permutations α, β, γ of D satisfy (5) for all distinct $x, y \in D$, with $q \geq 5$. Then $\alpha = \beta = \gamma = \chi$, where χ extends to an automorphism of the multiplicative group $\mathbb{F}_q^\times$.*

Proof. The map $(x, y) \mapsto (\alpha(x), \beta(y))$ sends Ω injectively to itself and hence bijectively to itself. The row with first coordinate x omits column x ; its image omits $\beta(x)$, whereas the complete output row with first coordinate $\alpha(x)$ omits $\alpha(x)$. Thus $\alpha = \beta =: h$.

Writing $x = ry$, equation (5) becomes

$$h(ry) = \gamma(r)h(y) \quad (7)$$

whenever $r, y, ry \neq 1$. Extend $\gamma(1) = 1$. If $r, s, rs \neq 1$, choose $y \in \mathbb{F}_q^\times \setminus \{1, s^{-1}, (rs)^{-1}\}$; this is possible since $|\mathbb{F}_q^\times| \geq 4$. Applying (7) twice gives $\gamma(rs) = \gamma(r)\gamma(s)$. The inverse case follows by choosing $y \in \mathbb{F}_q^\times \setminus \{1, r\}$ and comparing $h(y)$ with $h(rr^{-1}y)$. Hence γ extends multiplicatively.

Equation (7) also makes $h(x)/\gamma(x)$ constant on D , say $h(x) = A\gamma(x)$. Since both h and γ permute D , one has $AD = D$. But $AD = \mathbb{F}_q^\times \setminus \{A\}$ and $D = \mathbb{F}_q^\times \setminus \{1\}$, so $A = 1$. $\square$

Lemma 6.2 (Compatibility with translation). *Assume $q \geq 5$. If permutations $\alpha, \beta, \gamma, \delta$ of D satisfy both (5) and (6), then all four are restrictions of one field automorphism of $\mathbb{F}_q$.*

Proof. Lemma 6.1 gives $\alpha = \beta = \gamma = \chi$ for a multiplicative-group automorphism χ . Define $\eta(z) = 1 - \chi(1 - z)$ on D . Substitution of $x = 1 - z$ and $y = 1 - w$ in (6) yields $\eta(z)/\eta(w) = \delta(z/w)$. A second application of Lemma 6.1 gives $\eta = \delta =: \lambda$, where λ is another multiplicative-group automorphism. Thus χ is multiplicative both in the original coordinates and after translating 1 to 0. We now show that this forces a Frobenius power.

Since $\mathbb{F}_q^\times$ is cyclic, write $\chi(x) = x^m$ and $\lambda(x) = x^n$, with $1 \leq m, n \leq q - 2$ and both exponents coprime to $q - 1$. The definition of η says

$$(1 - z)^m + z^n = 1$$

for every $z \in D$, and the identity also holds for $z = 0, 1$. The polynomial $(1 - X)^m + X^n - 1$ has all q field elements as roots but degree at most $q - 2$, so it vanishes identically. Comparing degrees gives $m = n$, and then $(1 - X)^m = 1 - X^m$. All intermediate binomial coefficients vanish in characteristic p . Write $m = p^r u$ with $p \nmid u$. If $u > 1$, the coefficient of X^{p^r} in $(1 - X)^m = (1 - X^{p^r})^u$ is $-u \neq 0$, contradicting the vanishing of intermediate coefficients. Thus $u = 1$. Therefore all four maps are the same power of the Frobenius automorphism. $\square$

Proof of Theorem 1.1. For $q \geq 9$, Corollary 3.2 recovers the tangent traces, and Theorem 4.1 applies since $q - 2 > 4$. Thus every graph automorphism permutes the four recovered centre classes. Every permutation of a projective frame is realized by a unique projectivity. Compose by its inverse to obtain an automorphism preserving all four classes. In the normal form of Proposition 2.2, its symbol permutations satisfy (5) and (6); Lemma 6.2 makes them one field automorphism. This is induced by a semilinear collineation fixing the normalized frame.

Conversely, every semilinear frame stabilizer preserves legal points and collinearity, so it acts on G_K . The projective stabilizer of a frame is S_4 , realized in normalized coordinates over the prime field, and therefore commutes with Frobenius. The semilinear stabilizer is $S_4 \times C_e$, where $e = [\mathbb{F}_q : \mathbb{F}_p]$. Its action is faithful: an element fixing every legal point fixes each recovered trace, hence each centre. After normalization it is a field automorphism. Every $x \in D$ occurs as a first coordinate, so this field automorphism fixes D and also $0, 1$, and is the identity. $\square$

Proof of Corollary 1.2. Proposition 2.2 gives $|V(G_K)| = (q - 2)(q - 3)$, which is strictly increasing for $q \geq 3$. Hence $q = q'$. Choose a projectivity h carrying K to J . The map $h^{-1}f$ is a graph automorphism, so Theorem 1.1 supplies an extension. Uniqueness follows from the faithful action proved in Theorem 1.1. $\square$

Corollary 6.3 (Hamming-space extension). *Let $q = p^e \geq 9$, $D = \mathbb{F}_q \setminus \{0, 1\}$ and $Q = q - 2$. The code*

$$C_q = \{(x, y, x/y, (x - 1)/(y - 1)) : x, y \in D, x \neq y\} \subset D^4$$

has size $Q(Q - 1)$ and minimum Hamming distance 3. Restriction induces an isomorphism

$$\text{Stab}_{\text{Sym}(D) \wr S_4}(C_q) \longrightarrow \text{Isom}(C_q, d_H) \cong S_4 \times C_e.$$

Here the wreath product acts by permuting positions and independently permuting the symbols in each position.

Proof. Distinct words agree in at most one position, and a coordinate fibre has $q - 3 \geq 6$ words, so the minimum distance is three. The normal form counts $Q(Q - 1)$ words. Since distances between distinct words are three or four, code isometries are precisely graph automorphisms. Recovery of the four partitions extends each such automorphism to a position permutation and four symbol permutations. The four restricted partitions are distinct, and every symbol occurs in each position. An ambient Hamming isometry fixing every codeword therefore fixes the positions and all symbols, proving injectivity. Theorem 1.1 identifies the group. $\square$

The natural alphabet has Q symbols, which need not be a prime power. The Singleton bound for length four and distance three is Q^2 ; thus this code is not MDS. Hamming-space extension recovers position and symbol actions; Theorem 1.1 further identifies their projective semilinear origin.

Remark 6.4 (The small-order boundary). The division-table lemmas already apply for $q \geq 5$; the bound $q \geq 9$ enters only when the uncoloured graph is required to recover its four coordinate partitions. At $q = 5$, $G_K \cong K_6 \setminus 3K_2$, whose automorphism group has order 48, twice the projective frame stabilizer. Section 8 settles the remaining orders 7, 8 by exact finite enumeration.

7 Recognition and coordinate certificates

The proof can be made constructive without searching through all permutations of the field alphabet. First recover the four partitions. Then the missing entries in three partitions complete a division table, leaving only the possible images of a cyclic generator to test. The final check compares every pair of vertices, so the recovered coordinates are a certificate even when the input was not promised to come from geometry.

Lemma 7.1 (Recovering traces by common neighbours). *Let K be a four-point frame and B a tangent trace. A vertex outside B has at most three neighbours in B . Consequently, if Q consists of four vertices of B , then*

$$B = Q \cup \bigcap_{v \in Q} N(v).$$

For $q \geq 9$, the sets on the right that are cliques of size $q - 3$, as Q ranges over four-cliques, are exactly the tangent traces.

Proof. Write t for the centre of B . For an outside vertex z , a neighbour on B must lie on one of the three lines zu with $u \in K \setminus \{t\}$. Each such line meets the line of B at most once. Thus no outside vertex is adjacent to four vertices of B . Conversely all other vertices of B are adjacent to those four vertices. Every trace has size $q - 3$ by the normal form. Any resulting clique of that size has at least six vertices, so Lemma 3.1 places it in a tangent trace; equal sizes give equality. $\square$

Theorem 7.2 (Polynomial recognition over a supplied field). *Given a graph G with n vertices and a model of $\mathbb{F}_q$, $q \geq 9$, one can decide in polynomial time whether G is a frame continuation graph and, if it is, output a bijection*

$$\phi : V(G) \longrightarrow \{(x, y) \in (\mathbb{F}_q \setminus \{0, 1\})^2 : x \neq y\}$$

that preserves and reflects adjacency. A coarse bound for the procedure below is $O(n^6)$ elementary adjacency and field operations after rejecting $n \neq (q - 2)(q - 3)$. A supplied coordinate certificate can be checked using $O(n^2)$ field operations and adjacency tests.

Proof. Recover the partitions. Enumerate four-cliques and apply Lemma 7.1, retaining only clique closures of size $q - 3$, deduplicating equal sets. Reject unless there are $4(q - 2)$ traces. Construct their disjointness graph. Recover centre classes as the large maximal cliques from Theorem 4.1; concretely, enumerate five-cliques Q and form $Q \cup \bigcap_{B \in Q} N_{D_K}(B)$, again deduplicating equal sets. In a true frame graph every such five-clique is contained in one centre class, and a trace outside that class is disjoint from at most one member of it, so this closure is exactly the class. Reject unless there are four classes of $q - 2$ traces partitioning the recovered traces and each class partitions $V(G)$. These tests do not alone certify recognition.

Complete the division table. Order the four classes. The first two provide rows and columns. In a true frame graph each row meets each column in one vertex, except for one missing column; missing columns pair the row and column alphabets. Call their common alphabet D . Each row also omits exactly one symbol of the third class. Identify that missing symbol with its row label. In the normal form this is the identity $x/1 = x$: row x omits the quotient value x .

The third symbol at cell (x, y) now gives x/y for distinct $x, y \in D$. To complete division on $\mathbb{F}_q^\times$, we restore the identity row and column and the diagonal; zero does not belong to this table. Adjoin a symbol 1, put $x/x = 1$ and $x/1 = x$, and define $1/y$ as the unique symbol of D missing from column y . Set $1/1 = 1$. Reject inconsistent or nonunique entries. For a frame this completes the full division table of $\mathbb{F}_q^\times$.

For example, over $\mathbb{F}_5$ the punctured table is

x/y	2	3	4
2	—	4	3
3	4	—	2
4	2	3	—

Each row's missing column identifies its column label; the missing symbols in rows 2, 3, 4 are 2, 3, 4. The symbols missing from those columns are 3, 2, 4, giving the new identity row $1/2 = 3$, $1/3 = 2$, $1/4 = 4$. Fill the diagonal with 1 and the new identity column with the row labels. This example illustrates table completion only; recognition requires $q \geq 9$.

Assign field values and verify. Set $ab = a/(1/b)$. Try each element as a cyclic generator, checking its powers exhaust the $q - 1$ symbols and that the division table agrees with subtraction of exponents modulo $q - 1$. This recognizes a cyclic group directly, without assuming associativity of an unchecked table. Stop at the first successful cyclic-table witness; reject if none exists. Map this one generator to each primitive element of the supplied field in turn. For each map, form (x, y) coordinates and check that the fourth class is exactly the fibre partition of $(x - 1)/(y - 1)$. Finally check that the coordinates are a bijection onto the displayed set and that adjacency agrees for every pair. Output a passing map, or reject if none passes.

For a true frame any ordering of the four pencils is realized by a projectivity, so the completed table and at least one generator image pass these tests. Conversely the final check itself proves that a passing map is an isomorphism; no promised geometry is needed for soundness.

Cost. There are $O(n^4)$ four-vertex subsets, with $O(n^2)$ work per closure and clique check. After the trace-count check the disjointness graph has $O(\sqrt{n})$ vertices, so enumerating and checking its five-cliques costs $O(n^{7/2})$. Table completion, cyclic-table checks and all generator images take polynomial time bounded by $O(q^3 + qn^2)$ here. The initial vertex count gives $q = O(\sqrt{n})$, establishing the stated coarse bound. The last pairwise adjacency check is also the independent certificate verification procedure. $\square$

Proposition 7.3 (Size of the coordinate representation). *A frame graph has $(q - 2)(q - 3)$ vertices, degree $4(q - 4)$, and $2(q - 2)(q - 3)(q - 4)$ edges. Four field symbols per vertex determine adjacency by equality in one coordinate.*

Proof. Each coordinate fibre in the normal form has $q - 3$ vertices. Distinct fibres through a vertex meet only at that vertex, so its neighbours split into four sets of size $q - 4$. Count edges by summing degrees. $\square$

The coordinate representation therefore uses $\Theta(q^2)$ field symbols instead of $\Theta(q^3)$ explicit edges. This is a storage statement, not a speedup claim for every graph operation. In particular, enumerating all edges still has its output cost. The supplied reference implementation and timing protocol are described in the verification appendix.

8 The finite boundary

A *parallel class* is a partition of $V(G)$ into $q - 2$ cliques of size $q - 3$, called blocks. A *resolution* is an unordered set of four parallel classes whose blocks partition the edge set of G . Here parallelism means disjointness of the vertex sets of the blocks. Thus it specifies both a K_{q-3} -decomposition and its resolution into parallel classes. Blocks need not be maximal cliques. The tangent traces at the four frame points give the *geometric resolution*.

Proposition 8.1 (Exact small-order census). *For the indicated frame graphs the following table is exact. Here b counts all $(q - 3)$ -cliques, a counts parallel classes, and r counts resolutions.*

q	$ V(G_K) $	$ \text{Aut}(G_K) $	b	a	r
5	6	48	12	8	2
7	20	24	180	1352	1
8	30	144	48	16	2
9	42	48	28	4	1
11	72	24	36	4	1

At $q = 5, 8$ the two resolutions form one orbit, and the stabilizer of the geometric resolution is exactly the ambient semilinear group, of index two. At $q = 7, 9, 11$ every graph automorphism is ambient. The two resolutions share all twelve blocks at $q = 5$ and no blocks at $q = 8$.

Computational proof and completeness. The certificate bundle named in Appendix A enumerates all cliques of size $q - 3$, all exact covers of the vertex set by such cliques, and all exact covers of the edge set by the resulting partitions. At each recursive step an uncovered vertex or edge is chosen and every available covering block is explored. This enumerates each unordered cover once and terminates because the uncovered set strictly decreases. A second implementation enumerates fixed-size cliques directly instead of deriving them from maximal cliques. The two complete enumerations agree after sorting and comparing hashes. Their graph and clique constructions differ, but they share the exact-cover recursion described above.

Graph automorphism generators are computed with Sage’s graph algorithm. Their generated groups are explicitly enumerated and every permutation is checked against the edge set. An independent nauty computation gives the same full group orders, establishing the upper bound as well as the witnessed subgroup. The action on all resolutions is checked explicitly. Finally, the division-table lemmas apply for every $q \geq 5$: an automorphism stabilizing the geometric resolution, after a frame permutation, is Frobenius. Hence its stabilizer is precisely the ambient group. The bundle supplies an explicit representative of the other coset in each exceptional case; multiplying it by the ambient group describes every nonambient automorphism. $\square$

The exceptional symmetries can be displayed directly. At $q = 5$, G_K is the octahedral graph $K_6 \setminus 3K_2$. Interchanging the nonadjacent vertices $(3, 2)$ and $(4, 3)$ while fixing the other four is a graph automorphism. It sends the vertical trace $\{(3, 2), (3, 4)\}$ to $\{(4, 3), (3, 4)\}$ and fixes the vertical trace $\{(2, 3), (2, 4)\}$. The latter image is vertical and the former is not, so this map fails to preserve the geometric centre classes and is nonambient.

At $q = 8$, encode $\mathbb{F}_8 = \mathbb{F}_2[a]/(a^3 + a + 1)$ by binary coefficient integers. Number the legal pairs lexicographically from 0 to 29. The certificate's exceptional representative is the involution

$$(2\ 21)(3\ 6)(4\ 9)(5\ 29)(7\ 19)(8\ 27) \\ (10\ 25)(11\ 28)(12\ 24)(13\ 20)(16\ 23)(18\ 22),$$

with unlisted vertices fixed. It sends the tangent block $\{0, 1, 2, 3, 4\}$ to $\{0, 1, 6, 9, 21\}$, which is not a tangent trace. Its images of the four geometric parallel classes give the other resolution. Since it is an involution, it exchanges the two resolutions; they share no blocks. The bundle checks this permutation on every edge.

Together with Theorem 1.1, the census proves semilinear rigidity for every prime power $q \geq 5$ except $q = 5, 8$. This combined statement has a finite computational premise only at $q = 7$; the theorem for $q \geq 9$ does not.

For $q = 9, 11$, the last three census columns also follow without computation: every $(q - 3)$ -clique is a full tangent trace, and any $q - 2$ pairwise disjoint traces belong to one centre class since $q - 2 > 4$. Thus there are $4(q - 2)$ blocks, four parallel classes and one resolution.

At $q = 7$ there are 1352 individual clique partitions but only one compatible resolution into four parallel classes. Thus uniqueness of the full resolution can hold even when its individual candidate partitions are plentiful. At $q = 8$ the two-resolution orbit is exactly the obstruction to extending all graph automorphisms semilinearly.

9 Scope, a stronger complex, and open problems

The theorem reconstructs the four classes of tangent traces and proves that all graph automorphisms are ambient. It does not say that the binary graph alone reconstructs every line of the projective plane for a general arc, and it says nothing about the outcome or Sprague–Grundy value of an avoidance game.

A configuration-space interpretation. The space $M_{0,5}$ parametrizes ordered configurations of five distinct projective-line points modulo projectivities. With markings $(0, 1, \infty, x, y)$, the model in Proposition 2.2 identifies the vertex set with $M_{0,5}(\mathbb{F}_q)$. Up to reparametrizing the target, the four maps in (2) forget respectively y , x , 1 , and 0 . Thus the graph records fibre coincidence for four of the five forgetful maps to $M_{0,4}$. For comparison, Bruno and Mella prove $\text{Aut}(\overline{M}_{0,n}) = S_n$ for the compactified moduli space over $\mathbb{C}$ when $n \geq 5$ [4, Theorem 3]. This algebraic automorphism theorem does not constrain arbitrary permutations of the finite rational point set preserving a single graph relation.

Beyond pairwise compatibility. There is a stronger residual object,

$$\Delta_K = \{X \subseteq V_K : K \cup X \text{ is an arc}\}.$$

Its minimal nonfaces are the incompatible pairs whose joining line contains one selected point, together with the collinear triples whose line contains no selected point. Thus Δ_K retains line-incidence information that G_K discards. We do not pursue reconstruction from this stronger object here.

Open problems. The present arguments isolate two extremal parameters. For $k \geq 3$, let $m(k)$ be the maximum size of a clique not contained in one tangent trace, taken over k -arcs in finite projective planes of all orders. Lemma 3.1 gives the exact value five for the Desarguesian restriction of $m(4)$; the definition here also includes non-Desarguesian planes. Over the same class, let $r(k)$ be the maximum size of a disjointness clique of tangent traces using at least two selected centres. Set either parameter to zero if its defining class is empty. We proved

$$m(k) \leq k(k-2) + 1, \quad r(k) \leq k \binom{k-2}{2}.$$

Problem 9.1. Determine $m(k)$ or $r(k)$ exactly for an infinite family of k , or improve their orders of growth. Improvements can lower the corresponding sufficient recovery bounds.

Problem 9.2. Give a computation-free proof of the boundary in Proposition 8.1, especially the two-resolution phenomenon at $q = 8$ and uniqueness despite the many individual partitions at $q = 7$.

Problem 9.3. For larger arcs, determine natural hypotheses under which recovery of the tangent traces and their centre classes forces every graph isomorphism to extend semilinearly.

A Verification and reproducibility

The stable-range results have written proofs and use no finite census. No theorem in this draft is claimed to be Lean-formalized. The small-order census is an exact computer-assisted result, independently replayed using different graph and clique algorithms; its trust base includes Python, SageMath, nauty, and their arithmetic and graph implementations.

The public bundle contains `verification/boundary.json`, its SHA-256 manifest, the generator, the independent replay, and the coordinate-certificate checker. From the repository root, `make check` checks the statement/evidence links and replays the finite census and deterministic recognition tests. `make boundary` regenerates the census with SageMath; `make benchmark` records a fresh timing sample. The verification README specifies the field models, versions, exact domains, relabelling seeds and trusted boundaries.

The reference recognizer accepts a supplied field with integer-encoded elements and supplies prime-field models and explicit models of orders 8, 9, 16, 25. Stable-range tests use $q = 9, 11, 13, 16, 17, 19, 25$, including nontrivial Frobenius transports in the extension fields. The mathematical recognition theorem allows any supplied finite field. Checks include relabelled inputs, invalid coordinate certificates, deleted edges, and degree-preserving edge switches. Recorded timings compare reconstruction with generic graph isomorphism on the same relabelled inputs, excluding graph construction. They are single-run measurements, not a claim that reconstruction is faster.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author assumes responsibility for the manuscript and its accompanying artifacts; this first draft remains subject to author and external review.

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